

Title:

Physiological Signals to Detect Irregularities in Sleep Breathing

Introduction

Hypoapnea and Obstructive Apnea are sleep-related breathing disorders that interfere with normal respiratory function. Abnormal breathing patterns are caused due to events such as cough or snuffle.

This project techniques the waking recordings of four topics and uses machine learning algorithms to classify checking occasions.

Dataset Description

The dataset consists of overnight sleep recordings (approximately 8 hours each, from 5 participants).

The signals for each participant comprise the following:

- Nasal Airflow (32 Hz)
- Thoracic Movement (32 Hz)
- SpO₂ (4 Hz)
- Flow event annotations
- Sleep profile information

The flow event annotations tell us when breathing irregularity occurs.

Visualization

To analyze the signals, airflow, thoracic movement and SpO₂ signals were plotted for the duration of recordings.

We plotted annotated breathing events overlaid on the airflow signal to visually examine abnormal breathing episodes.

The visualizations were captured and saved as PDF for each of the individual participants.

Signal Preprocessing

We used a Butterworth bandpass filter (0.17–0.4 Hz) to filter the respiratory signals, such that the noise in the signal was removed but the range of breathing frequencies was retained.

The resulting filtered signals were segmented into 30-second windows with a 50% overlap.

The windows consist of 960 samples ($30\text{ s} \times 32\text{ Hz}$) each.

Window Labeling

Labels were assigned to each window based on overlap with annotated breathing events.

A window was labeled with the event type if more than 50% of the window overlapped with an event.

Normal. If otherwise the window was tagged as such.

This resulted in a total of 8800 labeled windows in the final dataset.

Each window includes two channels:

- Airflow signal
- Thoracic movement signal

Model Training

Breath patterns were classified using a 1D CNN.

We used Leave-One-Participant-Out Cross Validation (LOPO) to assess the model:

- Train the model on 4 subjects
- Keep one human participant and test the model on that

- Do this for all 5 participants

Results

Average results across all folds:

- Accuracy: 0.53
- Precision (macro): 0.19
- Recall (macro): 0.14

Due to the class distribution imbalance, normal and abnormal breathing events, low precision and recall were achieved with moderate accuracy.

Conclusion

In this project, we developed a completed pipeline for detecting abnormal breathings in sleep.

It consisted of signal visualization, preprocessing and then creating the datasets then finally classifying that dataset with a 1D CNN.

Class imbalance limits model performance, but the system represents a working approach for automated breathing event detection.